DISCHARGEIQ

Product, Safety, and Validation Overview

DischargeIQ takes the paperwork a patient is handed on the way out of the hospital and makes it readable. It rewrites the patient's own discharge document into plain language, answers their questions using nothing but that document, and separately reports what the document never explained so a care team can close the gap. Every clinical judgement stays with a human: the app explains what was already decided, and does not decide anything itself.

1. Why DischargeIQ Exists

Patients are discharged with paperwork written for clinicians and billing systems, then asked to manage their own recovery from it. Comprehension is low, and the documents themselves are frequently incomplete. Across our 106-document de-identified corpus, warning signs appear in only 34% of documents, discharge medications in 62%, and follow-up instructions in 69%.

- Patients often cannot tell which instruction is urgent and which is routine.
- Explanations must reflect the patient's actual document, not a generic template for their diagnosis.
- The system must communicate uncertainty clearly and never present AI output as medical advice.
- Gaps in the discharge document itself must surface to a human, not be silently filled by the AI.

2. What the App Does Today

DischargeIQ combines document extraction, plain-language explanation, and gap detection to support two primary use cases:

Patient Companion	Turns the discharge document into seven plain-language views: what happened, medications and why each was prescribed, recovery timeline, warning signs, follow-up appointments, a teach-back quiz, and a grounded chat that answers only from the document. Its purpose is to help a patient leave knowing what to do and what to watch for.
AI Patient Simulator	Reads the same document as a confused patient would and lists the questions the document does not answer, graded critical / moderate / minor with an overall gap score. Its purpose is to surface documentation gaps to a care coordinator or nurse before the patient is left alone with the paperwork.

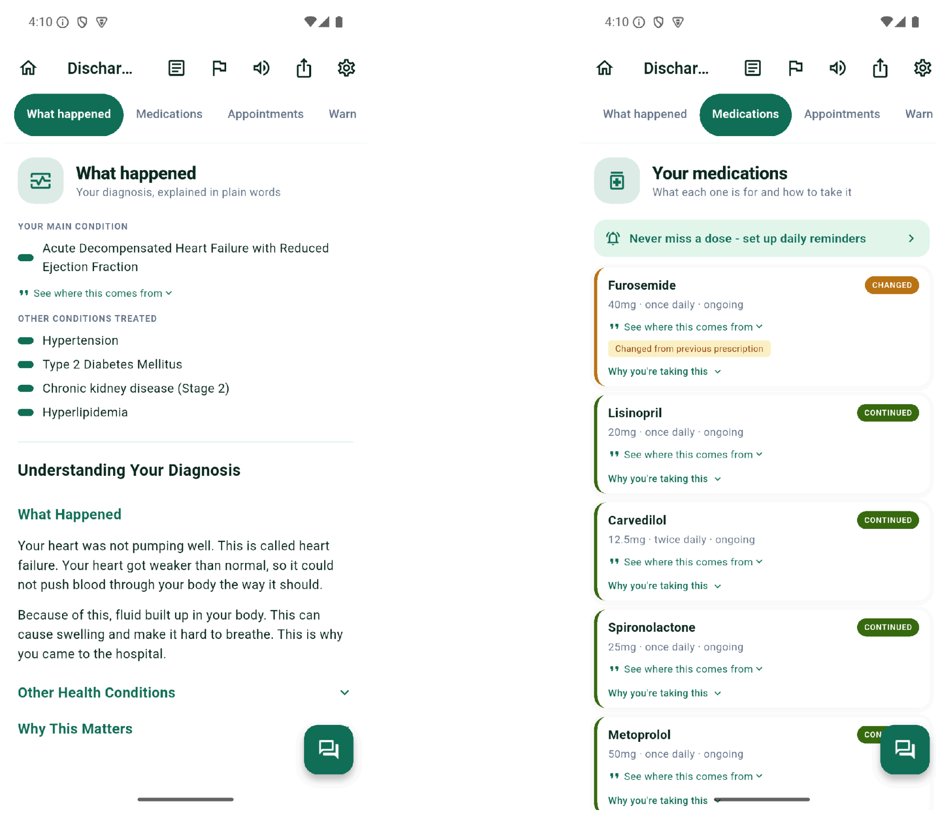

Figure 1. What happened, and your medications. Every extracted item carries a "see where this comes from" link that opens the exact source line. Medications are labelled new, changed, continued, or discontinued, and "why you're taking this" explains the reason without ever advising a dose change.

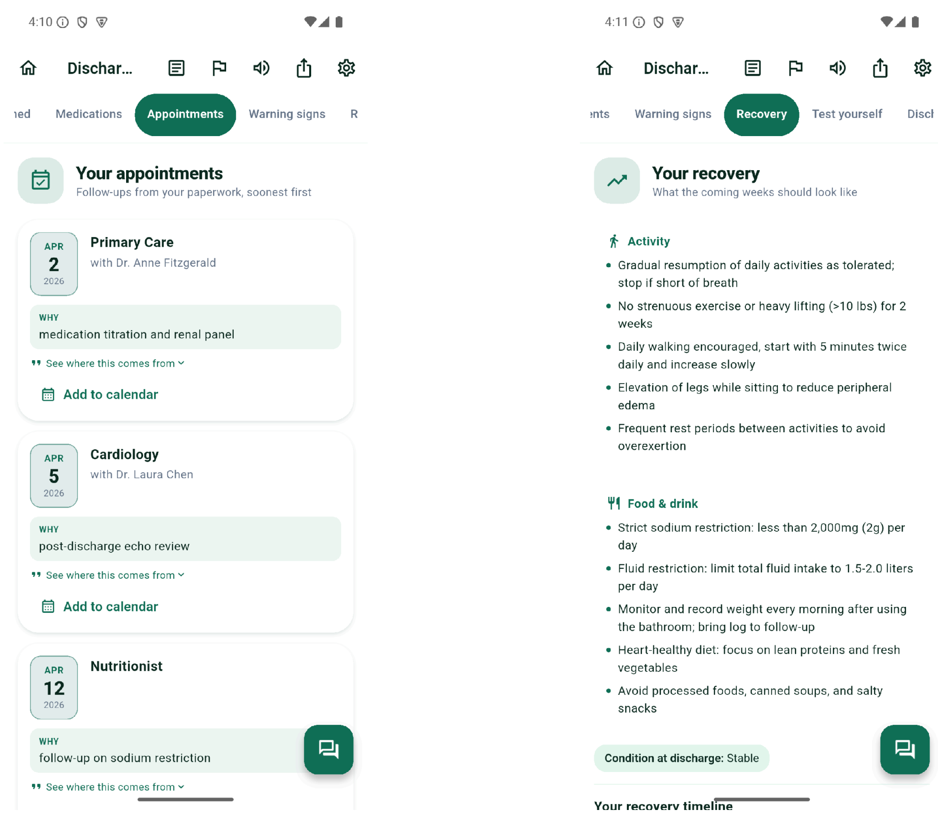

Figure 2. Your appointments, sorted soonest first with the stated reason for each and a one-tap calendar entry, and Your recovery, which separates activity limits from dietary limits and carries the condition recorded at discharge.

3. The Pipeline, End to End

1. The patient uploads or photographs their discharge document.
2. A router agent checks if the document is a discharge summary at all and rejects bills, explanation-of-benefit forms, and invoices before any clinical agent runs.
3. An extraction agent pulls structured fields - diagnosis, medications, appointments, restrictions, red-flag symptoms - and returns null rather than guessing any field the document does not contain.
4. Four explanation agents run in parallel over that structured extraction: diagnosis, medication rationale, recovery timeline, and a three-tier escalation guide.
5. A readability gate scores every output on Flesch-Kincaid and targets grade 6 or below.
6. The patient simulator agent re-reads the document and reports what it never answered.
7. The patient can take a five-question teach-back quiz before and after reading; the before/after delta is the comprehension measure we report.

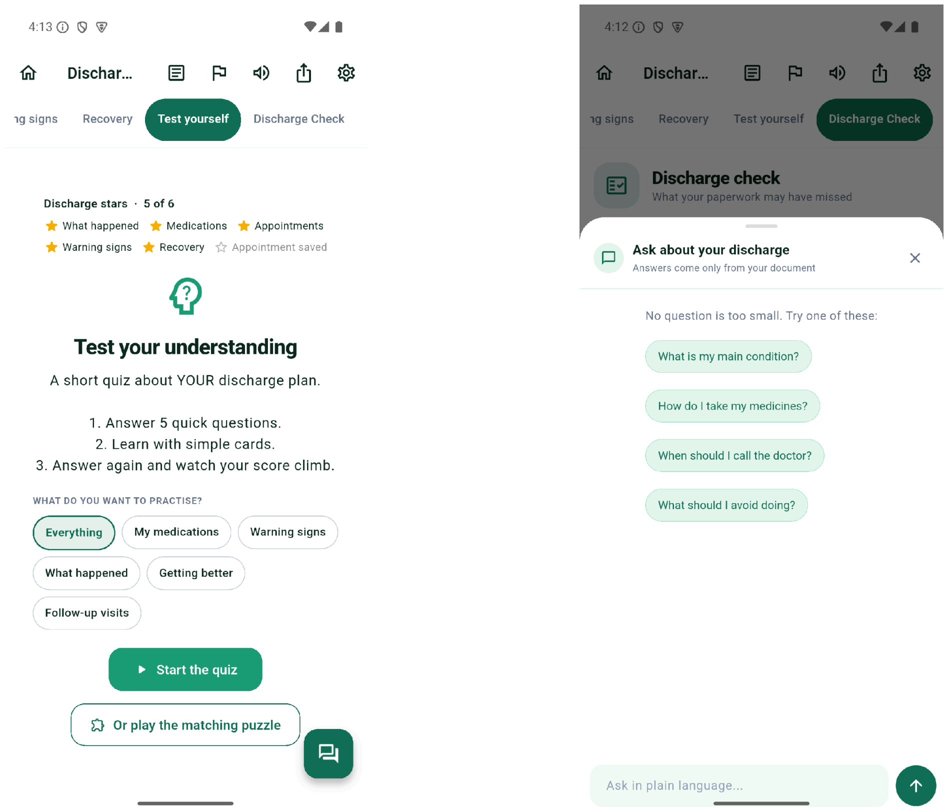

Figure 3. Test yourself, the teach-back quiz, taken once before reading with no hints or score shown and again afterwards - the delta is the comprehension figure we report. Beside it, the grounded chat, which states on its own header that answers come only from the patient's document.

4. Where Safety Lives in the Architecture

DischargeIQ is designed as more than a general-purpose chatbot. The prototype separates document extraction, clinical explanation, safety rules, and gap detection into distinct components, so that each can be inspected and constrained independently.

Extraction Contract	**A locked structured schema is the single contract between the document and every downstream agent. Fields absent from the document are returned as null. No downstream agent ever sees the raw document, only what was verifiably extracted from it.**
Grounding Check	Every medication, dose, and diagnosis in the output is checked back against the source text. Content that does not trace to the document is flagged rather than shipped.
Explanation Agents	Four specialised agents write the patient-facing sections from the extraction, each scoped to only the fields it needs and each blocked from clinical advice outside its remit.
Escalation Layer	Warning signs are presented in three tiers - call 911, go to the emergency department today, call your doctor - with a hard rule of zero ambiguous language. Every escalation output is read manually before it ships.
Gap Detection	The simulator agent scores what the document failed to explain and routes it to a human. The AI surfaces the gap; a clinician decides what to do about it.

Readability Gate	All patient-facing text is scored with Flesch-Kincaid against a grade 6 target, and failures send the prompt back for revision rather than reaching the patient.

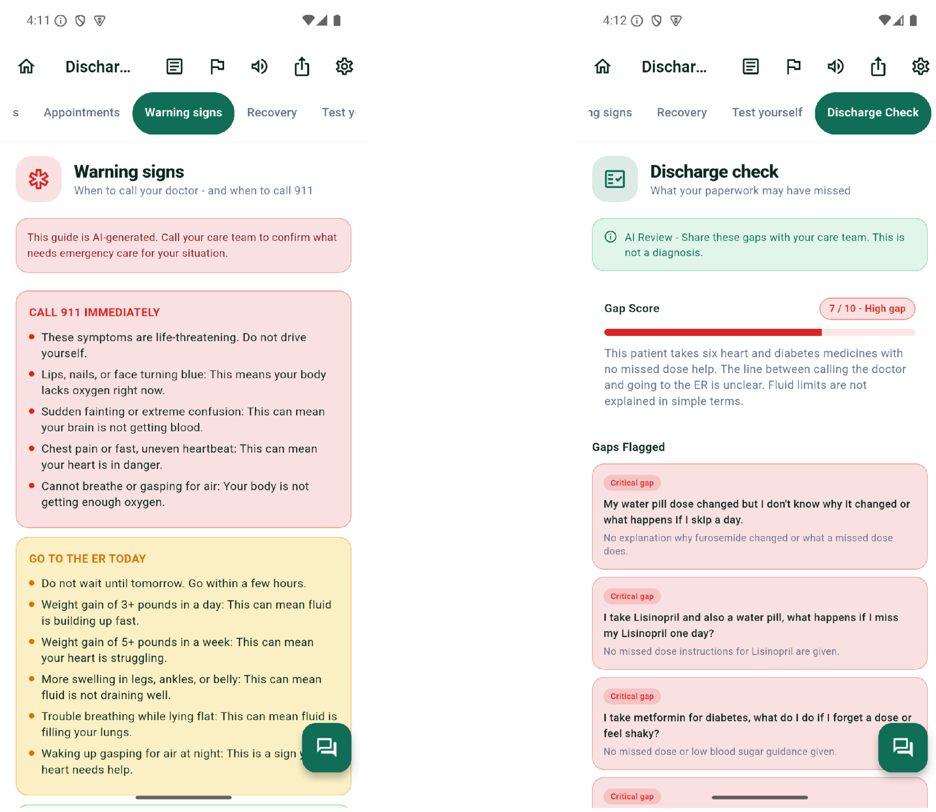

Figure 4. The two screens most in need of your review. Warning signs (left) apply the three-tier structure and opens with a standing caveat that the guide is AI-generated. Discharge check (right) is the gap report: a score, a plain summary, and each gap written as the question a patient would actually ask - addressed to the care team, labelled as not a diagnosis.

5. What the System Will Not Do

- DischargeIQ will not diagnose any condition, and will not interpret a symptom the patient reports.
- It will not prescribe, change, adjust, or recommend stopping any medication. It explains why a drug was prescribed and never touches what the patient should take.
- It will not answer from general medical knowledge. The chat answers only from the uploaded document and refuses everything else.
- It will identify red-flag situations and direct the patient toward emergency services or their care team rather than handling them conversationally.
- It will distinguish supportive explanations from medical advice, and label any content that did not come from the patient's own document.
- It will communicate uncertainty when a document is incomplete or ambiguous rather than filling the gap.

- Gaps it detects are framed as items to discuss with the care team, never as findings.

6. Evidence Base and How We Are Validating

Validation combines a structured clinician review of system output against source documents with a measured comprehension study. The purpose is to establish whether the plain-language output is faithful to the document it came from, and whether patients understand more after reading it.

- A 106-document corpus of real de-identified discharge documents drives testing, alongside a locked 50-document synthetic corpus for reproducible evaluation.
- Clinician reviewers score output for fidelity to the source document rather than against an ideal summary, because real discharge paperwork is itself incomplete.
- Any medication, dose, or diagnosis appearing in the output but not in the source is treated as the serious failure, and is checked automatically on every run as well as by reviewers.
- Comprehension is measured by a before-and-after teach-back quiz generated from the patient's own document, not from a question bank.
- No real patient data is used. Every test document is synthetic or de-identified.

WHAT INFORMED THE DESIGN

AHRQ Re-Engineered Discharge (RED) Toolkit

Used to inform which discharge elements matter most to patient understanding and to post-discharge safety. https://www.ahrq.gov/patient-safety/settings/hospital/red/toolkit/index.html

AHRQ Health Literacy Universal Precautions Toolkit - Teach-Back Method

Used as the basis for the teach-back quiz design and for the wording of the self-assessment rungs. https://www.ahrq.gov/health-literacy/improve/precautions/tool5.html

MTSamples de-identified transcribed medical reports

Source of the 106-document real-world discharge corpus used for extraction and hallucination testing. https://mtsamples.com/

7. Where the Build Stands, August 2026

Product scope	**Refined to a focused, pilot-ready patient-facing workflow: upload, seven plain-language views, grounded chat, and teach-back quiz.**
Software	End-to-end pipeline is implemented and deployed. Mobile application is the primary surface; a web demo surface runs the same backend.
Corpus	106 real de-identified documents plus a locked 50-document synthetic corpus, both wired into automated testing.
Clinician review	Review rubric and reviewer portal are built. Reviewer recruitment and the pass bar remain open and are the subject of question 5 below.
Patient validation	Teach-back quiz loop is shipped and instrumented. A patient-facing pilot has not started and depends on the governance answers in question 5.

8. Five Decisions I Need Clinical Input On

Below are the five decisions I cannot make without clinical authority. Each is a call I own as product lead, each is currently shipped one way, and each will change what we build depending on the answer. I have deliberately left out anything the team can settle itself, and anything about models, hosting, or architecture.

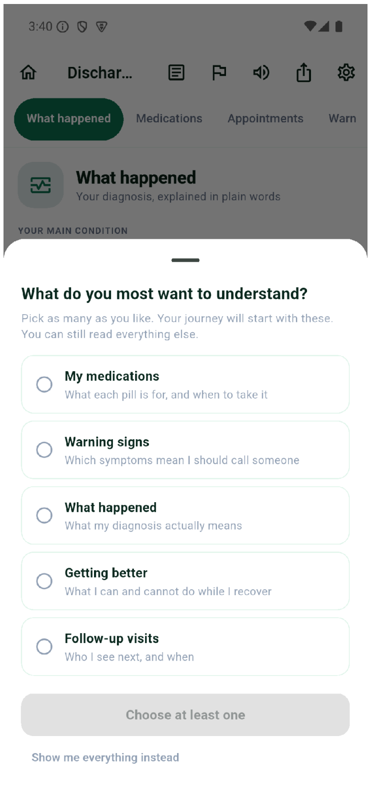

Figure 5. The screen behind questions 1 and 3: the patient chooses what they most want to understand, which reorders their reading. Nothing is hidden, and warning signs stay in the quiz whatever they pick.

1. Which parts of a discharge document should we explain first, because patients most often misunderstand them or come to harm from misunderstanding them?

The decision I own: the reading order of the seven tabs, and which topics the teach-back quiz weights most heavily.

Shipped today: all sections carry equal weight, and the patient can reorder them by choosing a topic they most want to understand. Warning signs are the one section never dropped from the quiz, whatever they choose.

What changes: if some content is more dangerous when misunderstood, it moves ahead of patient choice and joins the protected list. That list is one line of prompt text - the question is which topics belong on it.

2. What should the app show when a discharge document lists no warning signs at all - which is the case in roughly two-thirds of the real documents we hold?

The decision I own: whether to show a safety net the document did not provide, or to show nothing.

Shipped today: a short general emergency list - chest pain, trouble breathing, heavy bleeding, fainting, sudden weakness - labelled plainly as general advice and not from the patient's document. Where a document does list warning signs, they are presented in three tiers: call 911, go to the emergency department today, call your doctor.

What changes: if any non-document clinical content crosses a line, we ship an empty state and the 34% becomes a finding we report rather than a gap we fill. I would also value your read on whether three tiers is the right structure, or whether there is a symptom class where tiering is itself unsafe and the app should only say 911.

3. What patient and document information is genuinely essential for personalising the explanation, and what should we stop collecting?

The decision I own: what the app stores about a patient, and what it asks them for.

Shipped today: the document itself, an age band so that a summary belonging to a child is written to the caregiver rather than addressing the child as "you", the topic the patient says they want to understand, and an optional daily mood check-in that softens what the app asks of them on a bad day. Everything is stored on the phone.

What changes: anything you call unnecessary comes out. I would rather cut a feature now than defend the data collection at a privacy review later.

4. How should the app communicate uncertainty when the document is incomplete, ambiguous, or contradicts itself - and is refusal the right behaviour at 2am?

The decision I own: what the app says when it does not know, and what the chat does with a question it cannot answer.

Shipped today: the extraction omits rather than infers, so a field absent from the document stays empty. The chat answers only from the uploaded document; asks anything else, including an urgent symptom question, it refuses and redirects to the warning-signs section and the emergency line.

What changes: if refusal plus redirect is too passive for a sick person typing at 2am, that interaction needs to be designed rather than defaulted, and I need to know what it should do instead.

5. What evidence and success measures would you expect before this is safe to put in front of patients?

The decision I own: what the clinician review measures, where its pass bar sits, and what we call sufficient evidence to run a pilot.

Shipped today: reviewers score output for fidelity to the source document rather than against an ideal summary, on the reasoning that real discharge paperwork is itself incomplete. Content in the output that is not in the source is treated as the serious failure and is also checked automatically on every run. Comprehension is measured by a five-question quiz taken before and after reading.

What I need specifically: whether fidelity-to-source is the right instruction or absent content should count against us; where the pass bar belongs; whether a five-question before-and-after delta credibly stands in for teach-back as you practise it; and which approvals - IRB, legal, BAA, clinical governance - actually gate a patient-facing pilot, and in what order. A bar changed after scoring is not evidence, so I need this before the review starts, not after.

If it is more useful, I am happy to walk through the app live rather than on paper - the screens above are the ones the questions refer to. What I most want to hear is the answer I was not hoping for: the single thing that would make you stop trusting this output.

9. What This Is Not

Stating the limits plainly, because the questions above only make sense against them. DischargeIQ has never been used by a real patient. It has no regulatory clearance and is not a medical device, and nothing in it is a route to urgent care - a patient in trouble needs 911, not this app. Its comprehension numbers come from instrumented testing on our own corpus, not from a study, and should not be quoted as clinical evidence. It reads English only. It can explain a document well and still be explaining a document that was wrong to begin with, which is the failure mode I am least able to detect on my own. What I am trying to establish this quarter is narrow: is the output faithful to its source, and what would have to be true before a patient could safely be given it.